

PROJECT REPORT: ONLINE SHOPPING DATABASE SYSTEM

Technical architecture, ERD , and security implementation.

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PROJECT NAME : ONLINE SHOPPING

BUSINESS CASE : E – COMMERCE

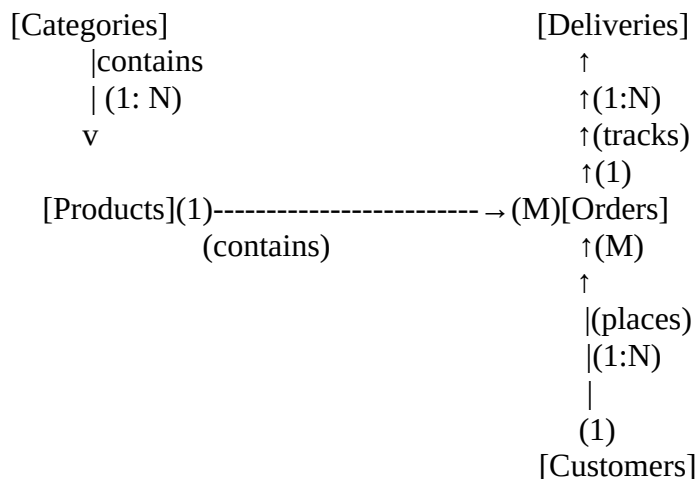
AREA OF INTEREST : SECURITY AND ACCESS CONTROL

EXECUTIVE SUMMARY

This project involves the design and implementation of an online shopping database. The system is designed to manage a full e-commerce lifecycle including inventory management , customer authentication , and order fulfillment. My specialised contribution focused on Security and access control , ensuring data privacy via cryptographic hashing and protecting the system integrity through Role Based Access Control (RBAC).

2. Entity-Relationship Diagram (ERD)

below is the database architecture reflecting the 5 core tables utilised in the system's operational lifecycle:



3. Database structure (DDL)

the database structure was implemented using SQL data Definition Language (DDL) to create normalized tables with appropriate indexing and relational integrity.

So my database includes five tables as shown below.

```
MySQL 8.0 Command Line Client - Unicode
Records: 2  Duplicates: 0  Warnings: 0

mysql> use onlineshopping;
Database changed
mysql> show tables;
+-----+
| Tables_in_onlineshopping |
+-----+
| categories                |
| customers                 |
| deliveries                |
| orders                    |
| products                  |
+-----+
5 rows in set (0.07 sec)

mysql> _
```

4. Security : Data Hashing implementation

I utilized the SHA2 algorithm to encrypt sensitive user passwords. This encryption to a 256_bit hash ensures that the original text is never visible in the database tables.

```
mysql> select * from customers;
```

CustomerID	Name	Email	Password	PhoneNumber	Address
10	John	john@gmail.com	89e01536ac207279409d4de1e5253e01f4a1769e696db0d6062ca9b8f56767c8	755454674	Kampala
11	John	john@gmail.com	89e01536ac207279409d4de1e5253e01f4a1769e696db0d6062ca9b8f56767c8	755454674	Kampala
12	Joshua	joshua@gmail.com	c9a08c872012cd27b3ab2dd9045c6e2dd51f88c1dfb6fccc173b986465ba28e7	755454364	Kibuli
13	John Doe	johndoe@gmail.com	66820f0586ee81f098f78f40cb5f3436a86a5022f059b0f97cb3edf2a180105c	0700000001	Kampala
14	Jane Smith	jane@gmail.com	66820f0586ee81f098f78f40cb5f3436a86a5022f059b0f97cb3edf2a180105c	0700000002	Entebbe
15	David Kim	david@gmail.com	86dae4363674d138be27fb50031a1588283d95bd4801f1b3e2bfff33dd1e58a5	0755588838	Jinja

```
6 rows in set (0.00 sec)
```

mysql>

5. Access Control and Authorization

I configured the role based security by creating specific users and the principle of giving privileges using GRANT and REVOKE commands.

6. Verification and testing

The security model was verified by logging in as a restricted user and attempting a prohibited action.

The system successfully successfully blocked this operation with an error.

Here I tested using the MYSQL workbench . Below is the verification of what happens if the restricted user attempted to delete data as the error 1142 shows up which means that access is denied for that specific user.

